

ReactJS Module 3: Hands-On: 1

Class Component Problems

Step 1: Create a file named index.html in a folder with the following code.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Module 3 - Class Components Problem</title>
7 </head>
8 <body>
9   <div id="app"></div>
10  <script src="https://unpkg.com/react@16/umd/react.development.js" crossorigin></script>
11  <script src="https://unpkg.com/react-dom@16/umd/react-dom.development.js" crossorigin></script>
12  <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
13 </body>
14 </html>
```

Step 2: Then add the following code within a script tag into the html right at the end of body

```
13 <script type="text/babel">
14   class App extends React.Component {
15     constructor(props) {
16       super(props);
17       this.state = { message: "Hello, World!" };
18       this.handleClick = this.handleClick.bind(this);
19     }
20
21     handleClick(e) {
22       console.log(this.state.message);
23     }
24
25     render() {
26       return <h1 onClick={this.handleClick}>{this.state.message}</h1>;
27     }
28   }
29
30   ReactDOM.render(<App />, document.getElementById("app"));
31 </script>
```

Step 3: Open index.html in a web browser and observe the output.

Hello, World

Step 4: Open the console and observe the output.

```
Download the React DevTools for a better development experience: https://fb.me/react-devtools  
You might need to use a local HTTP server (instead of file:///): https://fb.me/react-devtools-faq  
▲ You are using the in-browser Babel transformer. Be sure to precompile your scripts for production - https://babeljs.io/docs/setup/
```

Step 5: Comment out the call to super in the constructor save the file.

```
14  ... class App extends React.Component {  
15  ...   constructor(props) {  
16  ...     // super(props);  
17  ...     this.state = { message: "Hello, World!" };  
18  ...     this.handleClick = this.handleClick.bind(this);  
19  ...   }
```

Step 6: Open the console and observe the error

```
✖ ▶ Uncaught SyntaxError: Inline Babel script: 'this' is not allowed before super()  
3 |     constructor(props) {  
4 |       // super(props);  
> 5 |       this.state = { message: "Hello, World!" };  
   |       ^  
6 |       this.handleClick = this.handleClick.bind(this);  
7 |     }  
8 |  
   at n.buildCodeFrameError (babel.min.js:2)  
   at e.buildCodeFrameError (babel.min.js:2)  
   at e.ThisExpression (babel.min.js:11)  
   at e.s [as _call] (babel.min.js:18)  
   at e.i [as call] (babel.min.js:18)  
   at e.o [as visit] (babel.min.js:18)  
   at e.visitQueue (babel.min.js:18)  
   at e.visitSingle (babel.min.js:18)  
   at e.visit (babel.min.js:18)  
   at Function.s.node (babel.min.js:1)
```

Step 7: Revert changes made in previous steps and in the handleClick method, add a line to console.log this

```
21 .....handleClick(e)·{  
22 .....    console.log(this.state.message);  
23 .....    console.log(this);  
24 .....}
```

Step 8: Open the page and click on the h1 tag in the browser then open the console and see the output of console log

Hello, World!

► App {props: {...}, context: {...}, refs: {...}, updater: {...}, state: {...}, ...}

Step 9: Comment out the line in constructor that binds the handleClick method and save it

```
15 .....constructor(props)·{  
16 .....    super(props);  
17 .....    this.state = { message: "Hello, World!" };  
18 .....    // this.handleClick = this.handleClick.bind(this);  
19 .....}
```

Step 10: Click on the h1 on the page again as you will see it says cannot read property state of undefined. Which is why we have to bind methods in a class that deals with events

```
✖ ► Uncaught TypeError: Cannot read property 'state' of undefined  
    at handleClick (<anonymous>:27:24)  
    at HTMLUnknownElement.callCallback (react-dom.development.js:182)  
    at Object.invokeGuardedCallbackDev (react-dom.development.js:231)  
    at invokeGuardedCallback (react-dom.development.js:286)  
    at invokeGuardedCallbackAndCatchFirstError (react-dom.development.js:300)  
    at executeDispatch (react-dom.development.js:383)  
    at executeDispatchesInOrder (react-dom.development.js:408)  
    at executeDispatchesAndRelease (react-dom.development.js:3401)  
    at executeDispatchesAndReleaseTopLevel (react-dom.development.js:3410)  
    at forEachAccumulated (react-dom.development.js:3382)
```